

# TASK 5: Canine Wellness Dataset

## About Dataset

### Canine Wellness Classification Dataset (Synthetic, 10,000 Samples)

#### Overview

This synthetic dataset simulates a wide range of dog breeds and their health-related characteristics. It is designed for binary classification tasks, where the target variable is whether a dog is considered healthy ("Yes") or not healthy ("No").

The data was generated to reflect realistic distributions of age, breed sizes, weight, diet, and lifestyle factors that contribute to canine health. A simple rule-based logic was applied to create meaningful interactions between features and determine the target label.

#### Features

| Column                      | Description                                            |
|-----------------------------|--------------------------------------------------------|
| ID                          | Unique identifier                                      |
| Breed                       | Dog breed (15 common breeds)                           |
| Breed Size                  | Size category: Small, Medium, Large                    |
| Sex                         | Male or Female                                         |
| Age                         | Age in years (1–13)                                    |
| Weight (lbs)                | Dog weight in pounds                                   |
| Spay/Neuter Status          | Spayed, Neutered, or None                              |
| Daily Activity Level        | None, Low, Moderate, Active, Very Active               |
| Diet                        | Hard food, Wet food, Special diet, or Home cooked      |
| Food Brand                  | Brand of food (well-known or “Special” if home cooked) |
| Daily Walk Distance (miles) | Estimated distance walked daily                        |
| Other Pets in Household     | Yes or No                                              |
| Medications                 | Whether the dog is currently on medication             |
| Seizures                    | History of seizures: Yes or No                         |
| Hours of Sleep              | Daily hours of sleep (8–14)                            |
| Play Time (hrs)             | Average daily play time                                |
| Owner Activity Level        | Lifestyle of the owner                                 |
| Annual Vet Visits           | Number of vet visits per year (0–4)                    |

| Column                  | Description                           |
|-------------------------|---------------------------------------|
| Average Temperature (F) | Average local temperature             |
| Synthetic               | Flag indicating the data is simulated |
| Healthy                 | <b>Target:</b> Yes or No              |

## TASK 6: Diabetes Prediction Dataset

### About Dataset

The provided dataset appears to be related to diabetes and contains various biomedical measurements and patient characteristics. Here's a detailed description of the dataset columns:

- **ID:** A unique identifier for each record in the dataset.
- **No\_Patien:** Another identifier for the patient. It might be a patient number or record ID.
- **Gender:** The gender of the patient (F for Female, M for Male).
- **AGE:** The age of the patient in years.
- **Urea:** Urea level in the blood (likely measured in mg/dL or mmol/L). Urea is a waste product of protein metabolism and can indicate kidney function.
- **Cr:** Creatinine level in the blood (likely measured in mg/dL or  $\mu\text{mol/L}$ ). Creatinine is another waste product that indicates kidney function.
- **HbA1c:** Glycated hemoglobin, a measure of average blood sugar levels over the past 2-3 months (expressed as a percentage).
- **Chol:** Cholesterol level in the blood (likely measured in mg/dL or mmol/L). This typically refers to total cholesterol.
- **TG:** Triglycerides level in the blood (likely measured in mg/dL or mmol/L). Triglycerides are a type of fat in the blood.
- **HDL:** High-density lipoprotein cholesterol level (often called "good" cholesterol, measured in mg/dL or mmol/L).
- **LDL:** Low-density lipoprotein cholesterol level (often called "bad" cholesterol, measured in mg/dL or mmol/L).
- **VLDL:** Very low-density lipoprotein cholesterol level (measured in mg/dL or mmol/L).
- **BMI:** Body Mass Index, a measure of body fat based on height and weight (calculated as weight in kilograms divided by height in meters squared).
- **CLASS:** The class label indicating the diabetes status of the patient. The possible values seem to be:
  - N: Non-diabetic
  - P: Prediabetic
  - Y: Diabetic
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